Site report: Cambridge DELPH-IN activities

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DELPH-IN related Cambridge activities

- Guy Emerson: distributional semantics based on functions (Tuesday).
- Matic Horvat: realization and MT experiments using DMRS (below).
- Ewa Muszyńska: MEng project on chunking long sentences (below).
- Ongoing work with David Mott and Ping Xue on using ERG/PET for English to Controlled English.
- Attempt to talk about the 'real' philosophical basis of MRS: `www.cl.cam.ac.uk/~aac10/papers/iwcs2015-slides.pdf` (newish bits: 28–31 and 52–60)

Hierarchical Statistical Semantic Translation and Realization: Matic Horvat

- Integrating DELPH-IN resources (DMRS) into modern statistical machine translation (Hiero) for robust and accurate translation.
- Same framework used for robust statistical realization from DMRS into surface strings (Horvat et al. 2015 IWCS).
- We expect realization to be especially useful with broken or partial DMRS graphs (e.g. a result of a transfer-based MT system).
- Initial translation experiments on English-German pair, initial realization experiments using Wikiwoods corpus (English).

DMRS/sentence chunking

Marcellina has hired Bartolo as her counsel, since Figaro had once promised to marry her if he should default on a loan she had made to him, and she intends to enforce that promise. (5943)

Intuition: split this up before realization/parsing and then recombine (cf text simplification).

Marcellina has hired Bartolo as her counsel. (2)
since
Figaro had once promised to marry her. (7)
if
He should default on a loan she had made to him. (13)
and
She intends to enforce that promise. (1)

DMRS/sentence chunking

- Task: produce well-formed chunks that can a) be processed independently b) be recombined without loss of information.
- Here: split DMRS based on patterns: a) useful for realization b) (possibly) allows learning chunk boundaries for parsing preprocessor.
- Evaluated by splitting long DMRSs (from Wikiwoods), chunking associated strings, parsing chunks, realizing, recombining, checking match with original.
- Tough evaluation, aiming for high precision.
- Not usable yet but extended experiments are planned.